

Intelligence

Lecture 3

Development of IQ Testing

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Key Themes

- The Wechsler Intelligence Tests
- Raven's Progressive Matrices Tests
- Correlations between different IQ tests and between subtests on same test
- Types of validity to consider wrt. intelligence tests
- Uses and misuses of IQ tests



Glossary of Key Terms

Wechsler tests and related terms

- Wechsler Intelligence Tests**

A family of widely used, individually administered IQ tests (for adults and children) that use many different subtests to profile intelligence.

- WAIS (Wechsler Adult Intelligence Scale)**

The main Wechsler test for adults and older adolescents (approx. 16–90 years). Gives an overall IQ plus index scores (e.g. Verbal Comprehension, Working Memory, Perceptual Organisation, Processing Speed).

- WISC (Wechsler Intelligence Scale for Children)**

The Wechsler test designed specifically for school-age children (roughly 5–16 years).

- Subtest**

One of the individual tasks within a larger IQ test battery (e.g. Vocabulary, Block Design). Each subtest targets a particular aspect of cognitive ability.

- Verbal Comprehension**

WAIS index capturing understanding and use of language (e.g. defining words, explaining proverbs, answering general-knowledge questions).

- Perceptual Reasoning**

WAIS index measuring non-verbal reasoning and visual–spatial problem solving (e.g. Block Design, Matrix Reasoning, Visual Puzzles).

- Working Memory**

WAIS index measuring the ability to temporarily hold and manipulate information in mind (e.g. Digit Span, Arithmetic, Letter–Number Sequencing).

- Processing Speed**

WAIS index measuring how quickly and accurately simple information can be processed (e.g. Coding, Symbol Search, Cancellation).

- Deviation IQ**

IQ defined in terms of how far a person's score deviates from the average score for their age group (mean = 100, SD = 15), rather than using mental-age/chronological-age ratios.

- Standardisation / Norming**

The process of giving a test to a large, representative sample (varying in age, sex, social class, region, etc.) to create norms that individual scores can be compared against.

Raven's Progressive Matrices

- Raven's Progressive Matrices (RPM)**

A non-verbal IQ test where people identify the missing piece in a pattern of shapes. Designed to measure abstract reasoning and to be relatively independent of language and culture.

- Abstract reasoning**

The ability to detect patterns, relationships and rules between shapes, ideas, or pieces of information, beyond concrete, everyday content.

- Eductive ability**

Raven's term for the ability to “draw out” meaning from complex or confusing information and see underlying patterns.

- Reproductive ability**

The ability to store, recall, and reproduce learned information.

Correlations, reliability and validity

- Correlation**

A statistical measure (ranging from -1 to $+1$) describing how strongly two variables move together. Positive correlations between subtests mean people who score high on one tend to score high on others.

- Hierarchy of mental abilities**

The idea that test scores can be organised in levels: individual subtests → clusters or “index scores” (e.g. Verbal Comprehension) → a general factor “g” at the top.

- Reliability**

How consistently a test measures something.

- Internal reliability**

The extent to which items within a test or subtest correlate with each other, suggesting they measure the same underlying construct.

- Test–retest reliability**

The extent to which a test gives similar results when the same person takes it on different occasions (assuming no real change in their ability).

- Validity**

Whether a test actually measures what it claims to measure (here: intelligence), and whether it predicts relevant outcomes.

- Face validity**

The extent to which a test appears, on the surface, to measure what it claims to (e.g. an IQ test that “looks like” it measures reasoning and problem-solving).

- Concurrent validity**

When a test correlates well with other established measures of the same construct (e.g. WAIS scores correlating with other IQ tests).

- Predictive validity**

The degree to which test scores forecast future outcomes such as school achievement, job performance, or training success.

Uses and misuses of IQ tests

- Educational use**

Using IQ tests to help identify children who may need extra support or who are exceptionally able, to plan appropriate educational provision.

- Clinical / diagnostic use**

Using IQ tests as part of an assessment for intellectual disability, learning difficulties, or the cognitive impact of brain injury, dementia, etc.

- Occupational use**

Using cognitive ability tests as one component in selection, placement, or training decisions in workplaces.

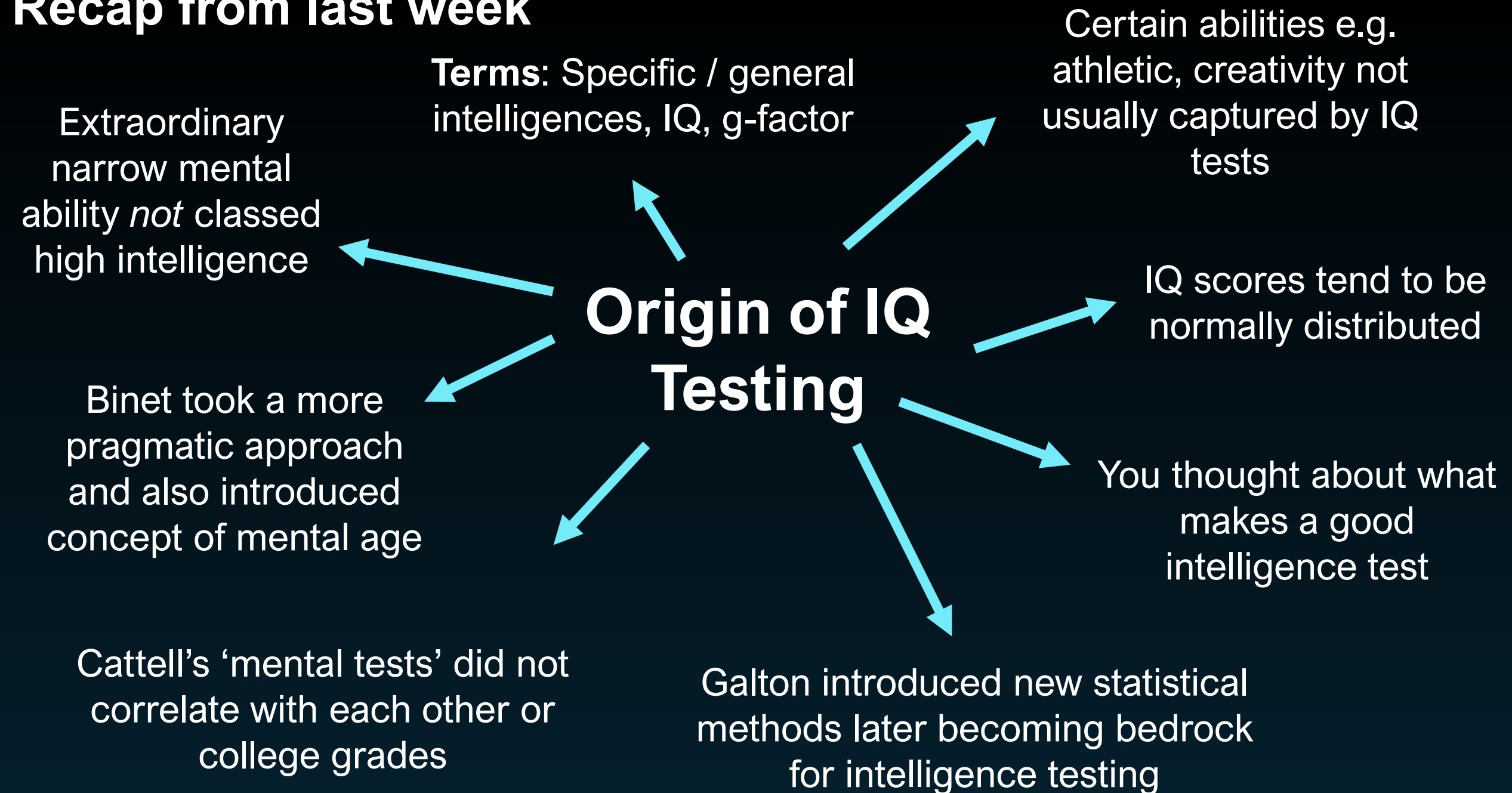
- Cultural bias**

The criticism that some IQ tests are more familiar or appropriate for people from certain cultural or educational backgrounds, which can disadvantage others.

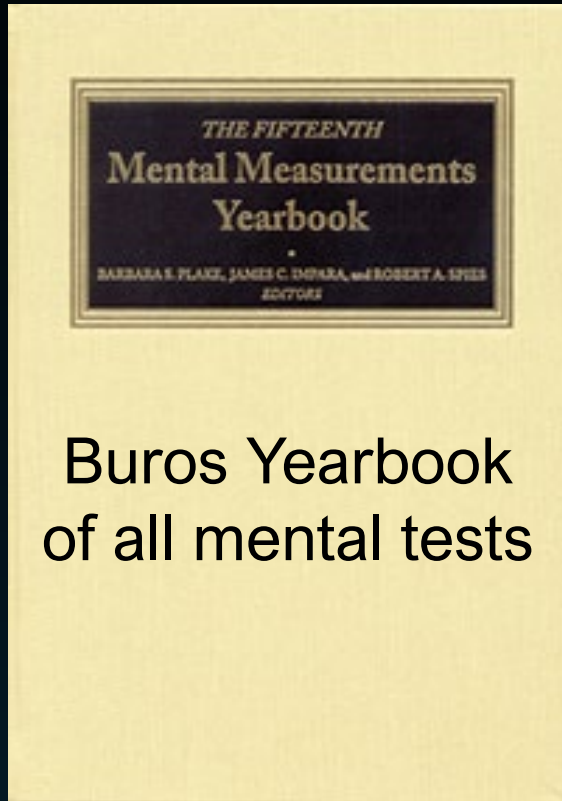
- Eugenics**

A movement (especially in the early 20th century) that misused IQ tests to justify policies like forced sterilisation, immigration restriction, and racial discrimination.

Recap from last week

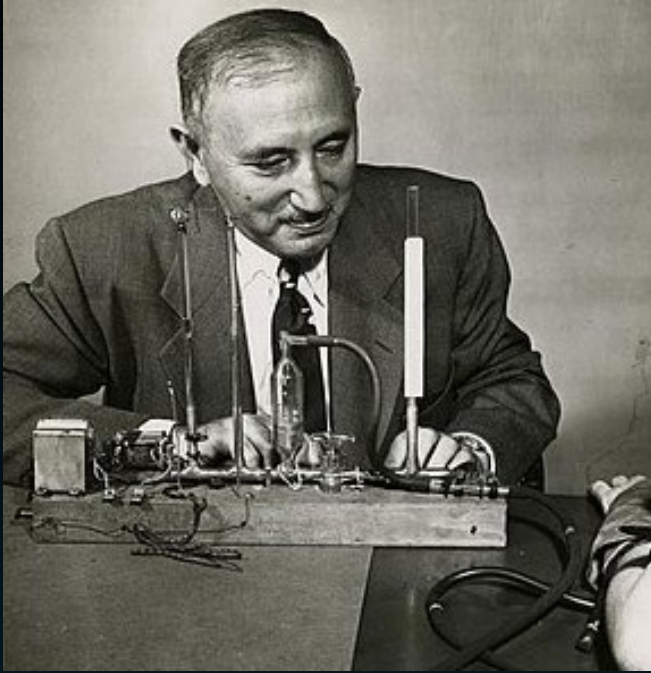


1920s and 1930s – great expansion of IQ tests



- Especially in US – great expansion in IQ tests – lots of rival tests
- Buros started publishing a yearbook, listing all mental tests, e.g. 1978 edition included 100 intelligence / mental tests

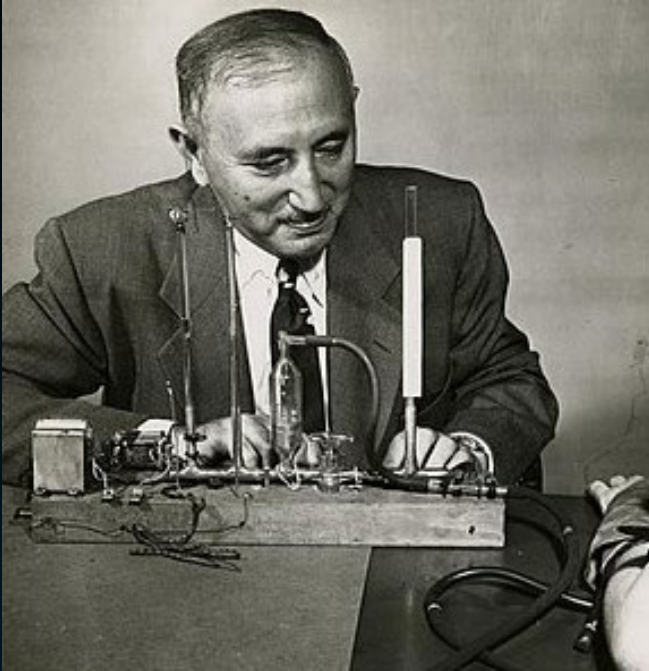
The Wechsler Tests



David Wechsler
1896-1981

- 1939 Wechsler-Bellevue Scale. Revised in 1955 Wechsler Adult Intelligence Scale
- Taken over the Stanford-Binet test as the most widely used IQ test today
- 2 versions:
 - *Wechsler Adult Intelligence Scale (WAIS)* – latest edition (IV) standardised among 2200 adults (16-90 years)
 - *Wechsler Intelligence Scale for Children (WISC)* – for children (5-16 years)

The Wechsler Tests



David Wechsler
1896-1981

- Created to profile his patients in the clinic
- Wechsler was sceptical that the Stanford-Binet relied heavily on verbal ability and only used a single score
- His test tapped into range of cognitive abilities (reflecting his belief there were many aspects to intelligence)
- *WAIS-IV* released in 2008 with fifth edition released in 2024

The Wechsler Adult Intelligence Scale

→ WAIS-R (1981) Subtests measuring different aspects of intelligence (including 6 verbal and 5 performance tests):

Verbal

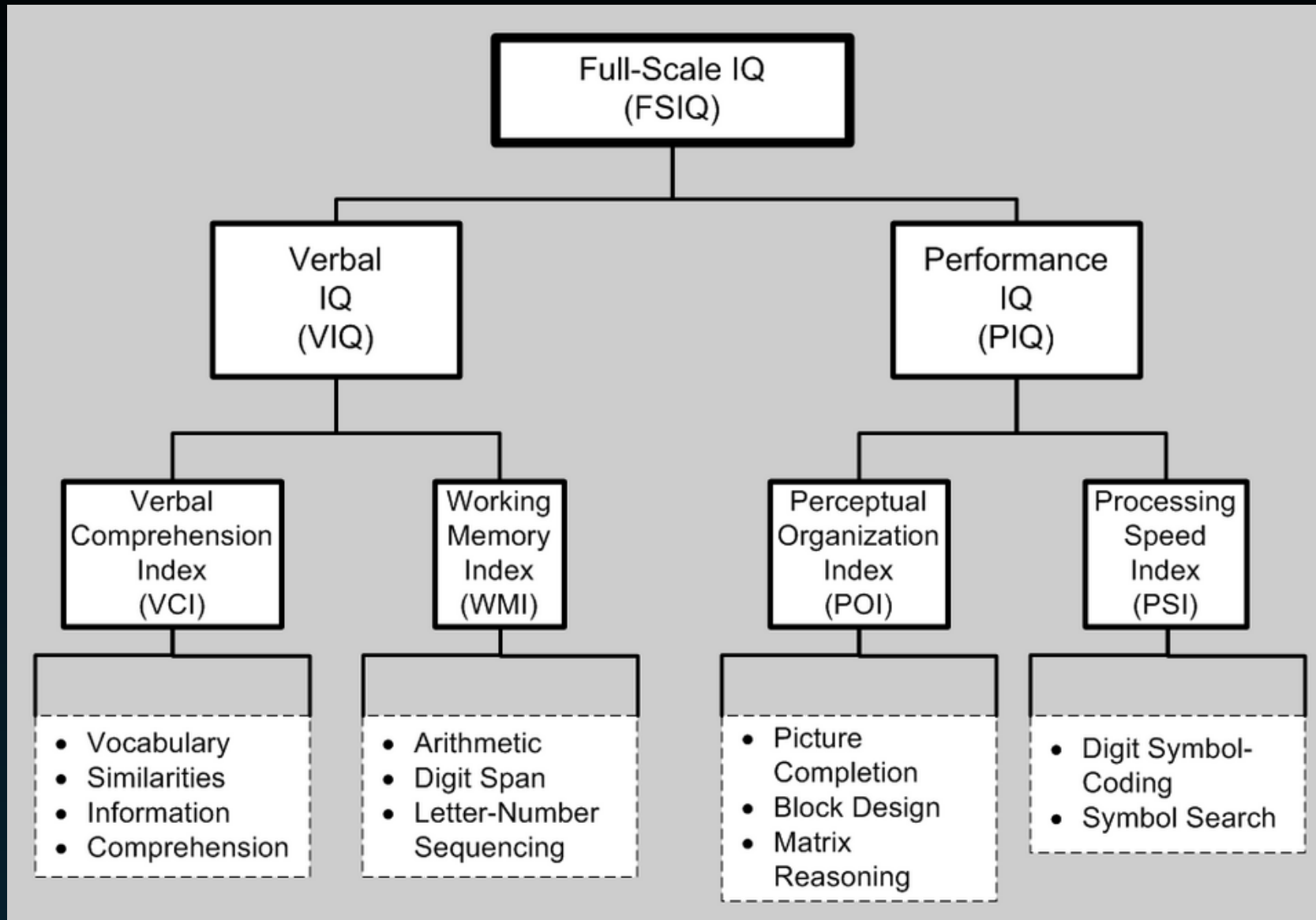
Information
Comprehension
Arithmetic
Digit Span
Similarities
Vocabulary

Performance

Picture arrangement
Picture Completion
Block Design
Object Assembly
Digit Symbol (perceptual speed)

A **verbal** and **performance IQ** score are obtained

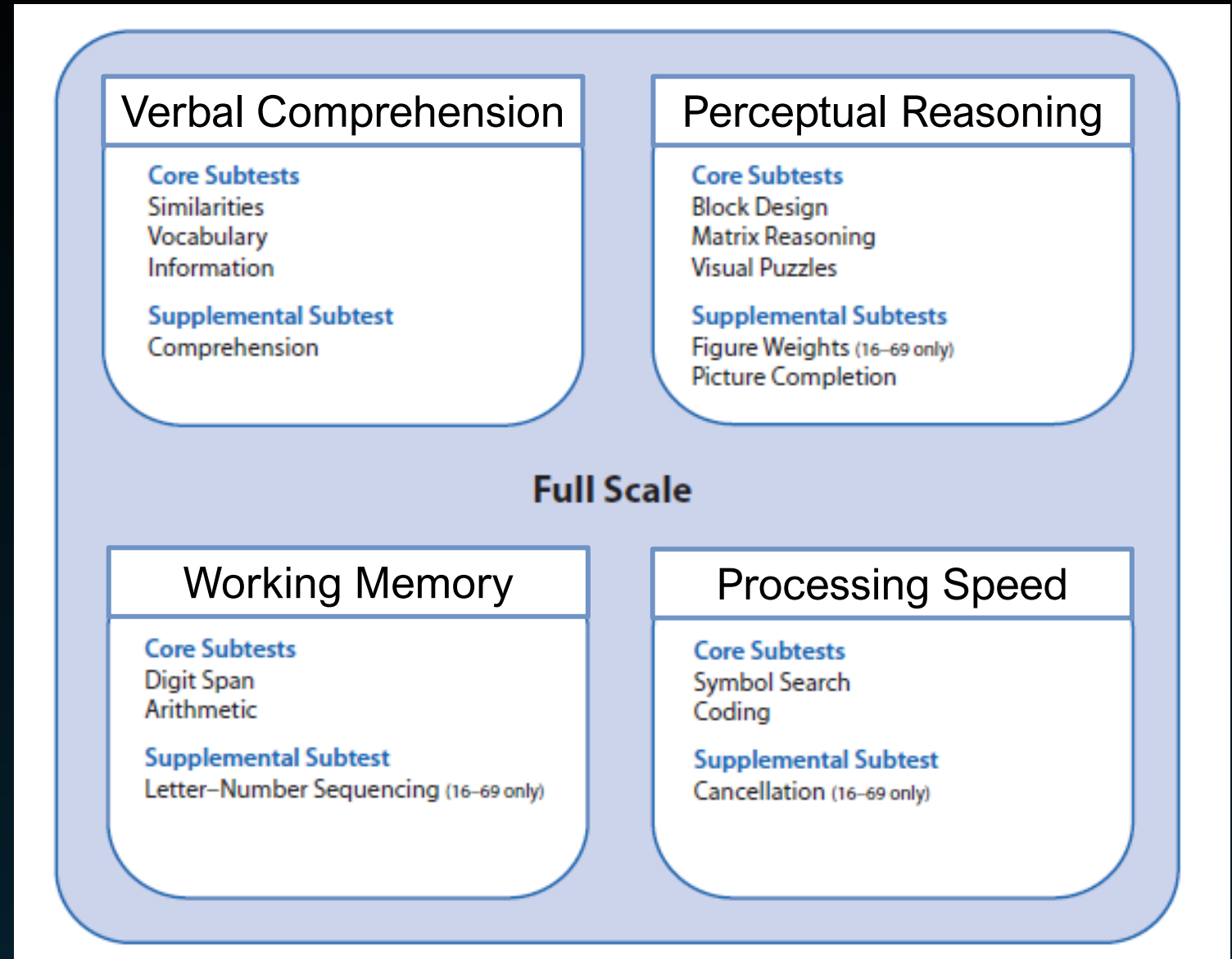
The Wechsler Adult Intelligence Scale (WAIS-III)



1997

The Wechsler Adult Intelligence Scale IV (2008)

- Widely used today, though WAIS-5 (2024) being unrolled
- 15 different mental tests (wide range of mental effort).
- Many hundreds of pounds to buy
- Can only be bought by certified people, e.g. educational, clinical, occupational psychologists
- Administered by trained psychological tester working one to one for up to 2 hours



For 16-90 year-olds

The Wechsler Adult Intelligence Scale IV

To sit the WAIS-IV test, you would have to complete the following 15 subtests:

Similarities

Say what two words have in common. For example: In what way are an apple and a pear alike? In what way are a painting and a symphony alike? How are whales and lions similar? (18 questions)

The Wechsler Adult Intelligence Scale IV

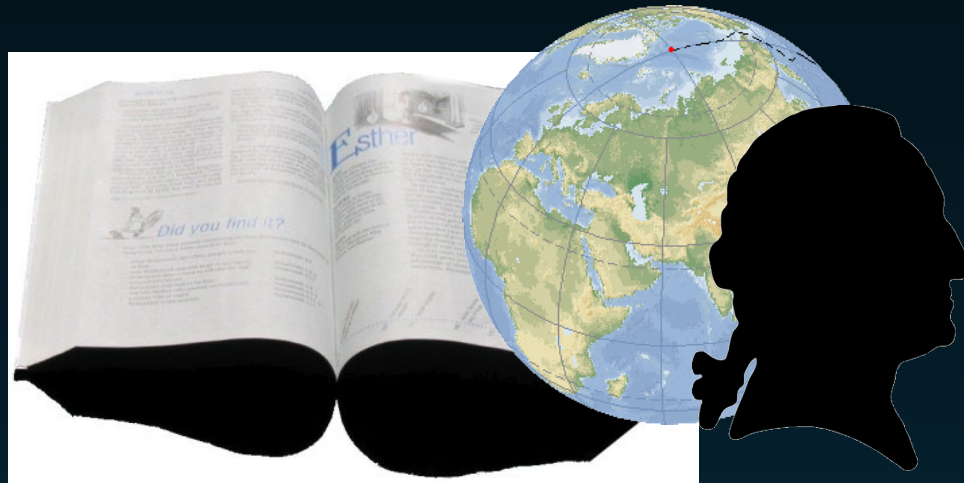
Vocabulary

Tell the tester what certain words mean. For example: chair (easy), hesitant (medium), presumptuous (hard). (30 words)

The Wechsler Adult Intelligence Scale IV

Information

General knowledge questions covering people, places, and events. For example: How many days are in a week? What is the capital of France? Name three oceans. Who wrote *The Iliad*? What imaginary line divides the planet earth into north and south hemisphere?
(26 questions)

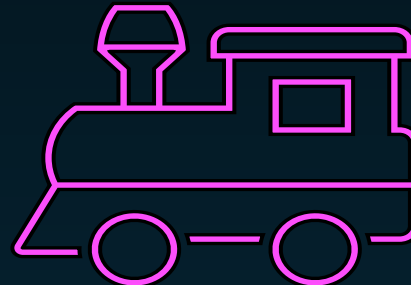
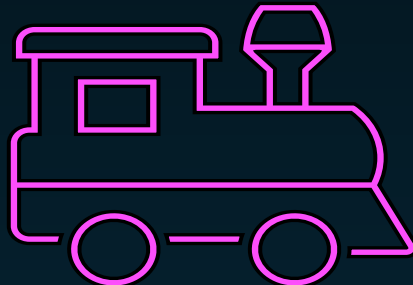


The Wechsler Adult Intelligence Scale IV

Comprehension

Answer questions about everyday-life problems, aspects of society, and proverbs. For example: Tell me some reasons why we put food in a refrigerator. Why do people require driving licences? What does it mean to say 'a bird in the hand is worth two in the bush'? (18 questions)

Example: 2 trains are travelling in opposite directions 100 miles apart; 1 train is travelling at 40 miles while the other train is travelling at 60 miles an hour. How long before the trains meet?

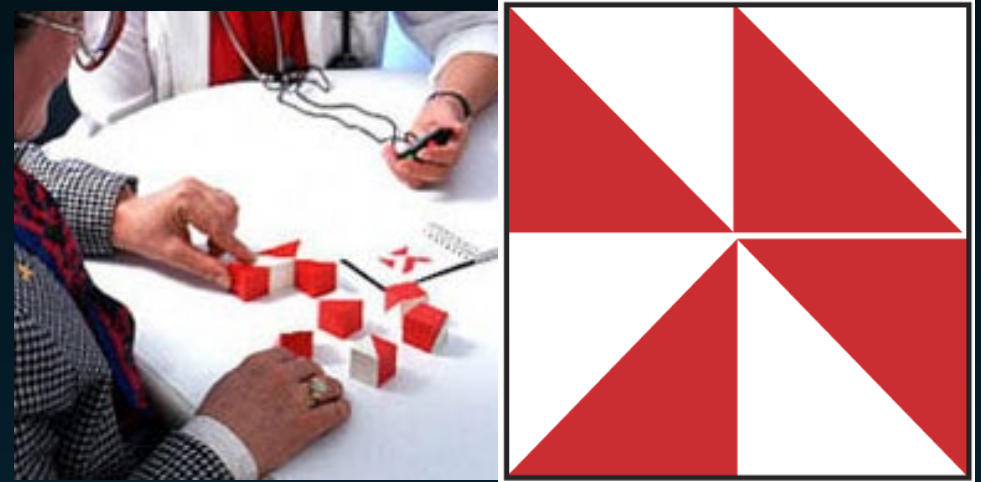


The Wechsler Adult Intelligence Scale IV

Block Design

While looking at a two-dimensional pattern made up of red and white squares and triangles, try to reproduce the pattern using cubes with red and white faces, and faces that are diagonally half-red and half-white.

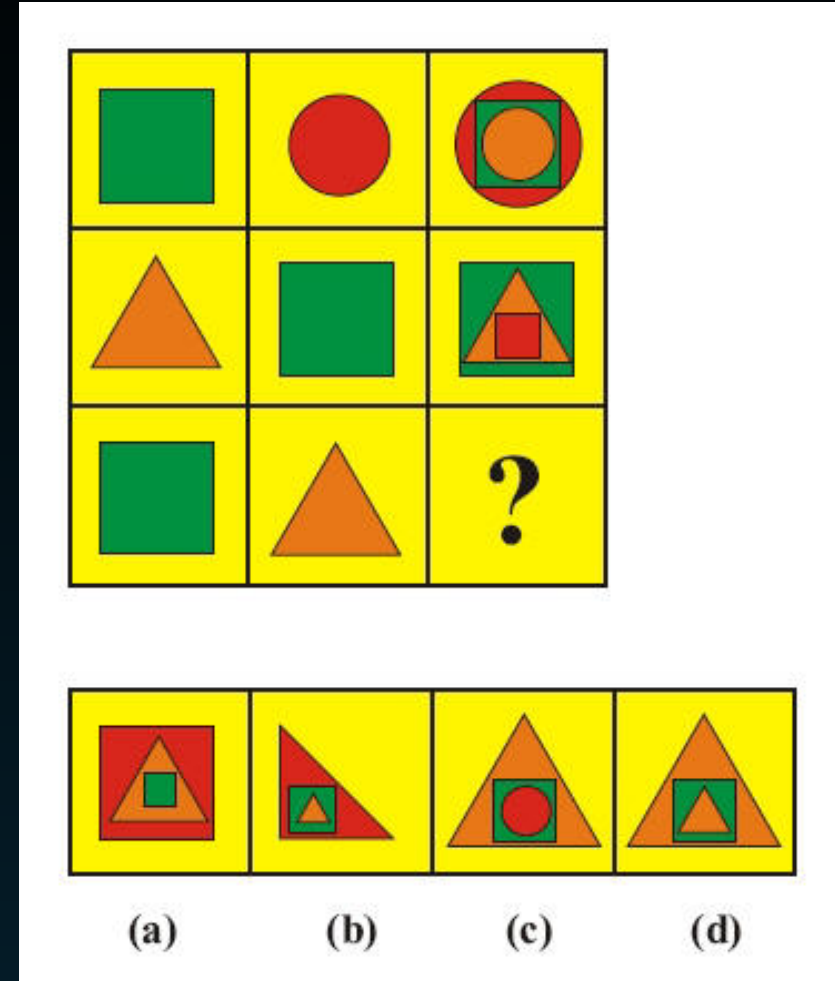
(14 patterns)



The Wechsler Adult Intelligence Scale IV

Matrix Reasoning

Find the missing element in a pattern that is built up in a logical manner. A fairly easy example of this type of task is shown right. (26 questions)

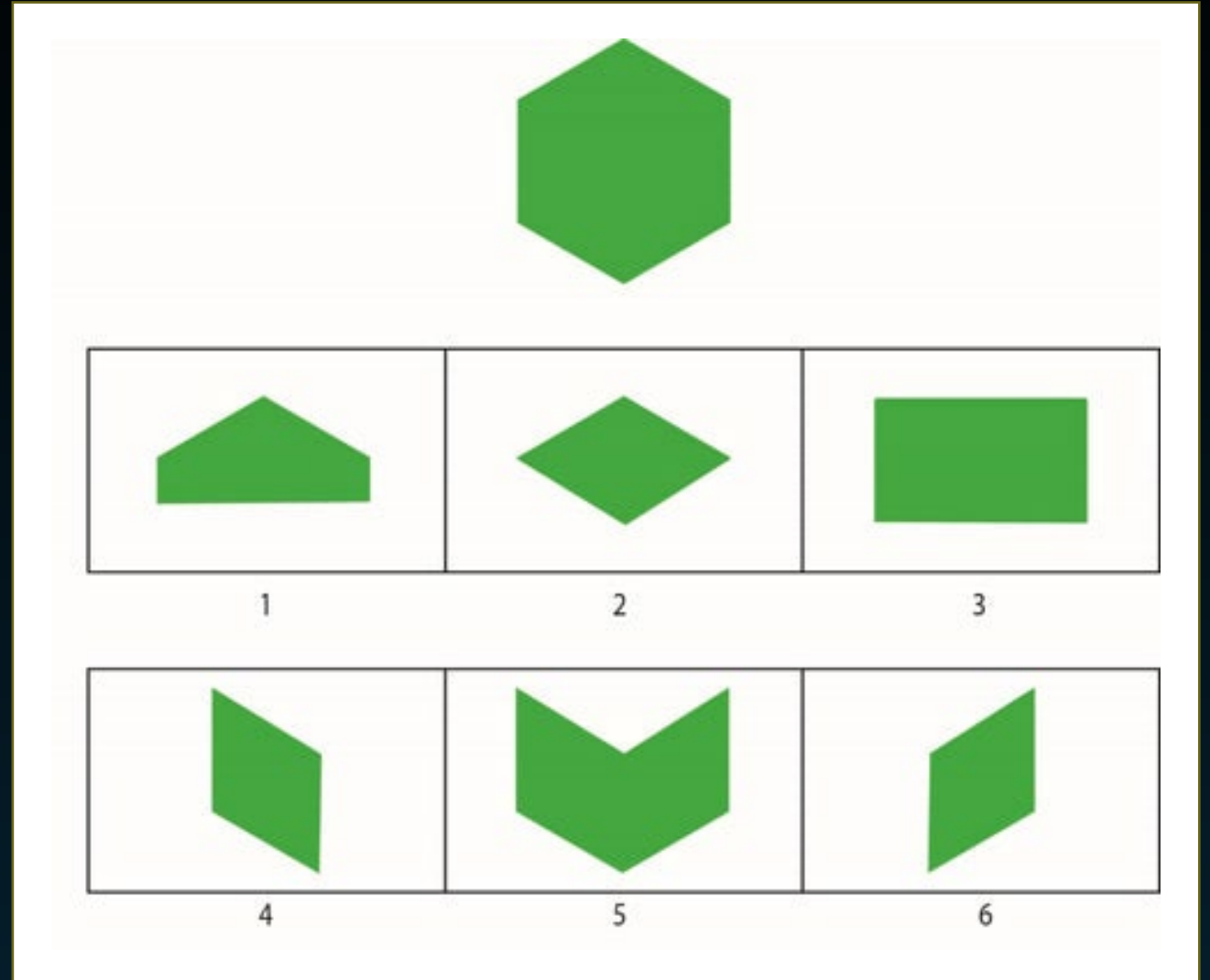


The Wechsler Adult Intelligence Scale IV

Visual Puzzles

There is a shape at the top of each page. Below it are six part-shapes. Find the three part-shapes that can be put together to make the shape at the top of the page.

(26 questions)

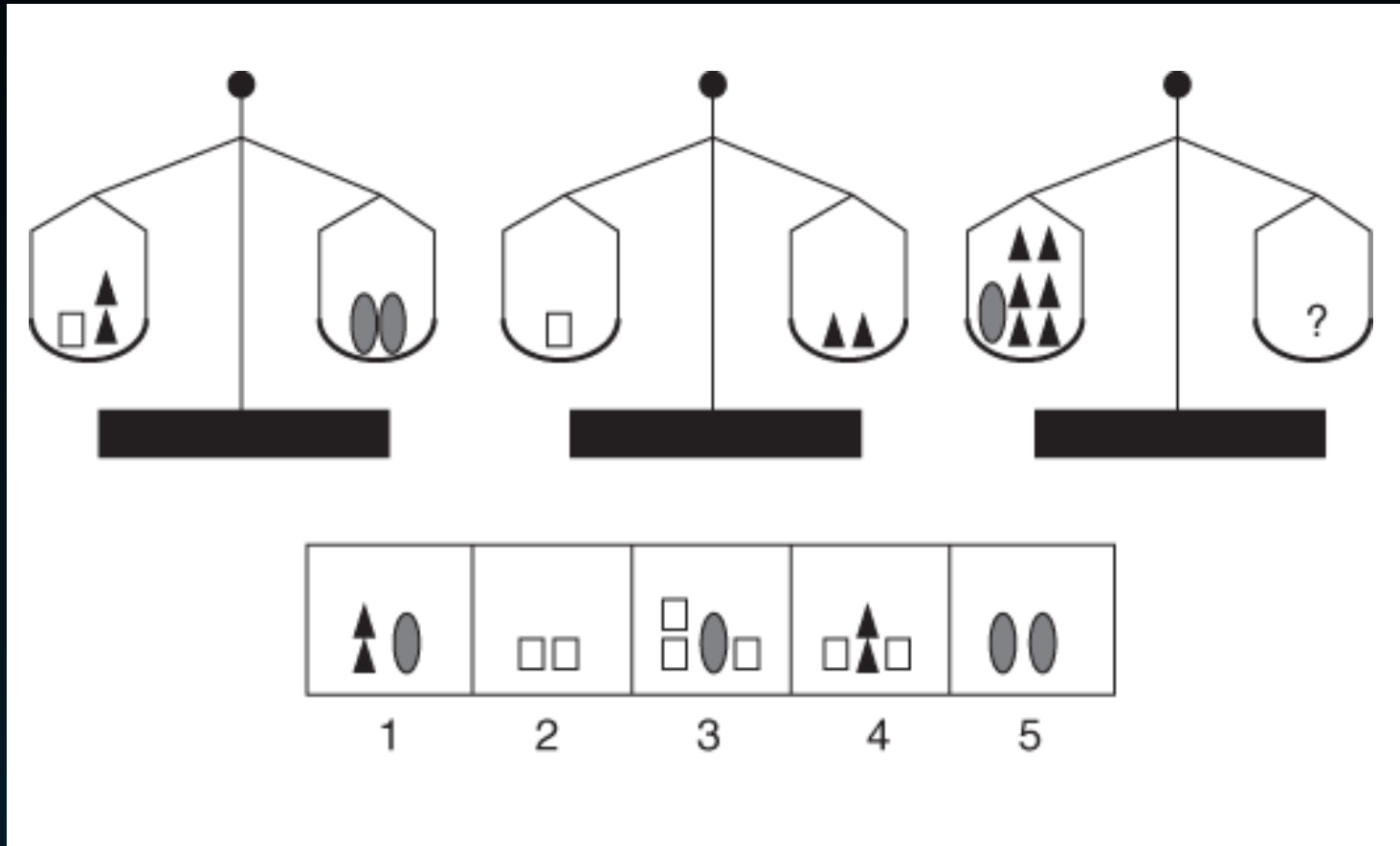


The Wechsler Adult Intelligence Scale IV

Figure Weights

Choose the correct objects to balance a scale in weight. A medium-hard example of this type of task is shown right.

(27 questions)



The Wechsler Adult Intelligence Scale IV

Picture Completion

Spot the missing element in a series of colour drawings. For example: that spokes are missing from one wheel in a picture of a bicycle; that one buttonhole is missing from a jacket in a picture of a person. As in the earlier tests in the collection, the questions become progressively more difficult. (24 drawings)



The Wechsler Adult Intelligence Scale IV

Digit Span

Repeat a sequence of numbers to the examiner. Sequences run from 2 to 9 numbers in length. An easy example is to repeat 3–7–4; harder is 3–9–1–7–4–5–3–9. In the second part of this test the sequences must be repeated in reverse order, e.g. 24-3-7-12 backwards is 12-7-3-24
(Maximum of 16 forward and 16 reversed sequences)

The Wechsler Adult Intelligence Scale IV

Arithmetic

Solve mental arithmetic problems. These involve simple counting, addition, subtraction, multiplication, division, and percentages.

(22 questions)

For example: *If you need 31 minutes to make two cakes, how much time do you need to make 12 cakes?*

The Wechsler Adult Intelligence Scale IV

Letter–number Sequencing

The examiner reads a list (as short as two, or as long as eight) of alternate letters and numbers. The person being tested must repeat them, putting the numbers first and in numerical order, followed by the letters in alphabetical order. (Maximum of 30 trials)

For example: ‘W-4-G-8-L-3’

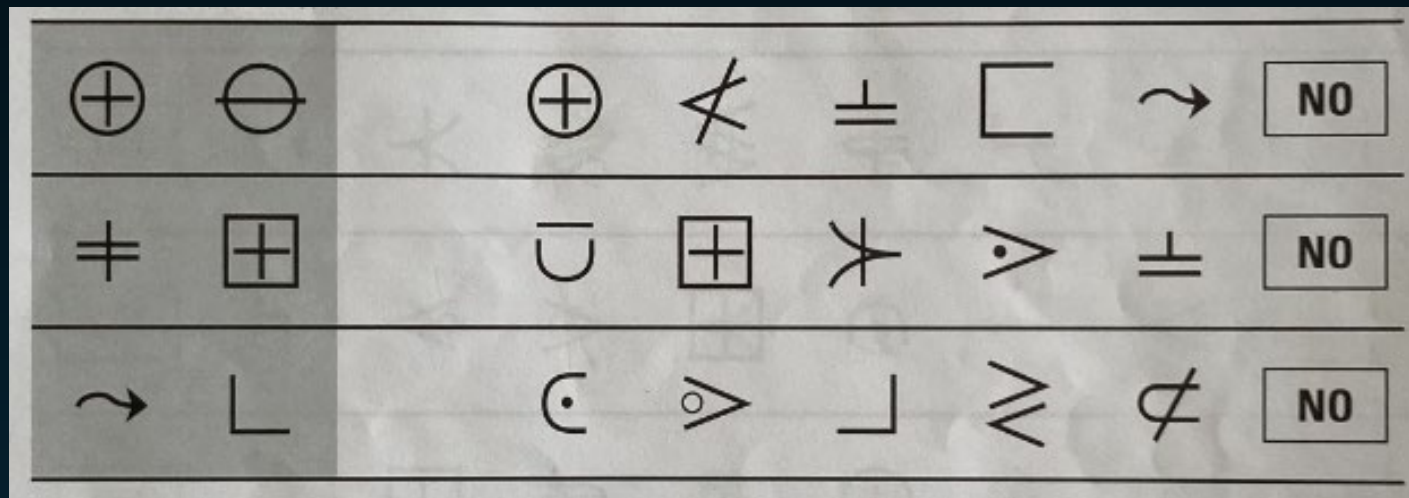
‘3–4–8-G-L-W’

The Wechsler Adult Intelligence Scale IV

Symbol Search

Identify, from a list of abstract symbols, which symbol in a given pair of target symbols is contained in the list. Assesses information processing speed and visual perception.

(As many as can be completed correctly in two minutes)



The Wechsler Adult Intelligence Scale IV

Coding

Write down the symbol that corresponds to a given number. An example of this type of task is shown right.

(As many as can be completed correctly in two minutes)

Key

1	2	3	4	5	6	7	8	9
>	—	≠	□	×		└	┐	▽

Practice

4	8	9	1	2	6	3	5	7

Test

3	2	5	6	9	1	2	7	7

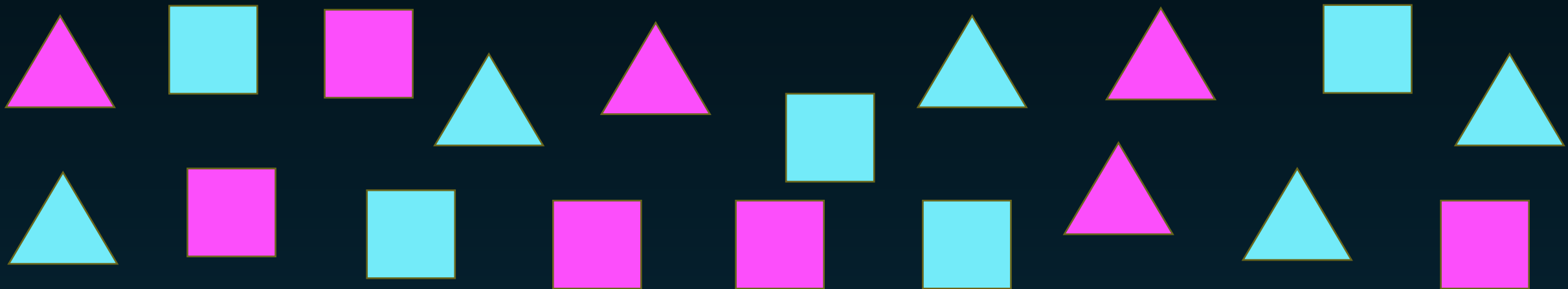
4	6	7	2	1	9	8	8	3

2	3	8	5	6	4	8	3	7

The Wechsler Adult Intelligence Scale IV

Cancellation

In a large sheet of paper with pink triangles and squares and blue triangles and squares, put a pencil line through each blue square and each pink triangle. There is a second sheet with red or yellow stars and circles. (The time to finish each sheet is taken)



The Wechsler Adult Intelligence Scale IV

seeing similarities and differences

drawing inferences

working out and applying rules

working out how to construct shapes

Mental Functions

working with numbers, attending to details

articulating the meaning of words

explaining practical actions in everyday life

processing simple information at speed

recalling general knowledge

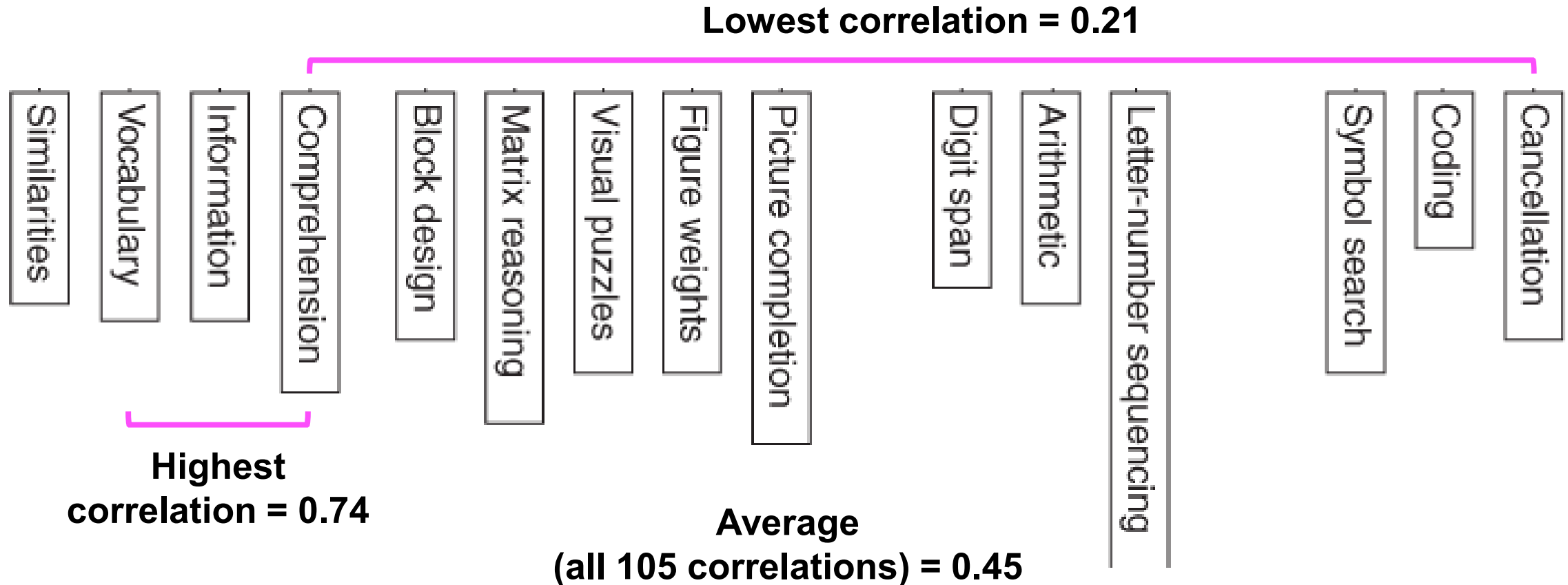
remembering and manipulating mental material

Question!

What do you expect to see in the relations (correlations) between the 15 different tests?

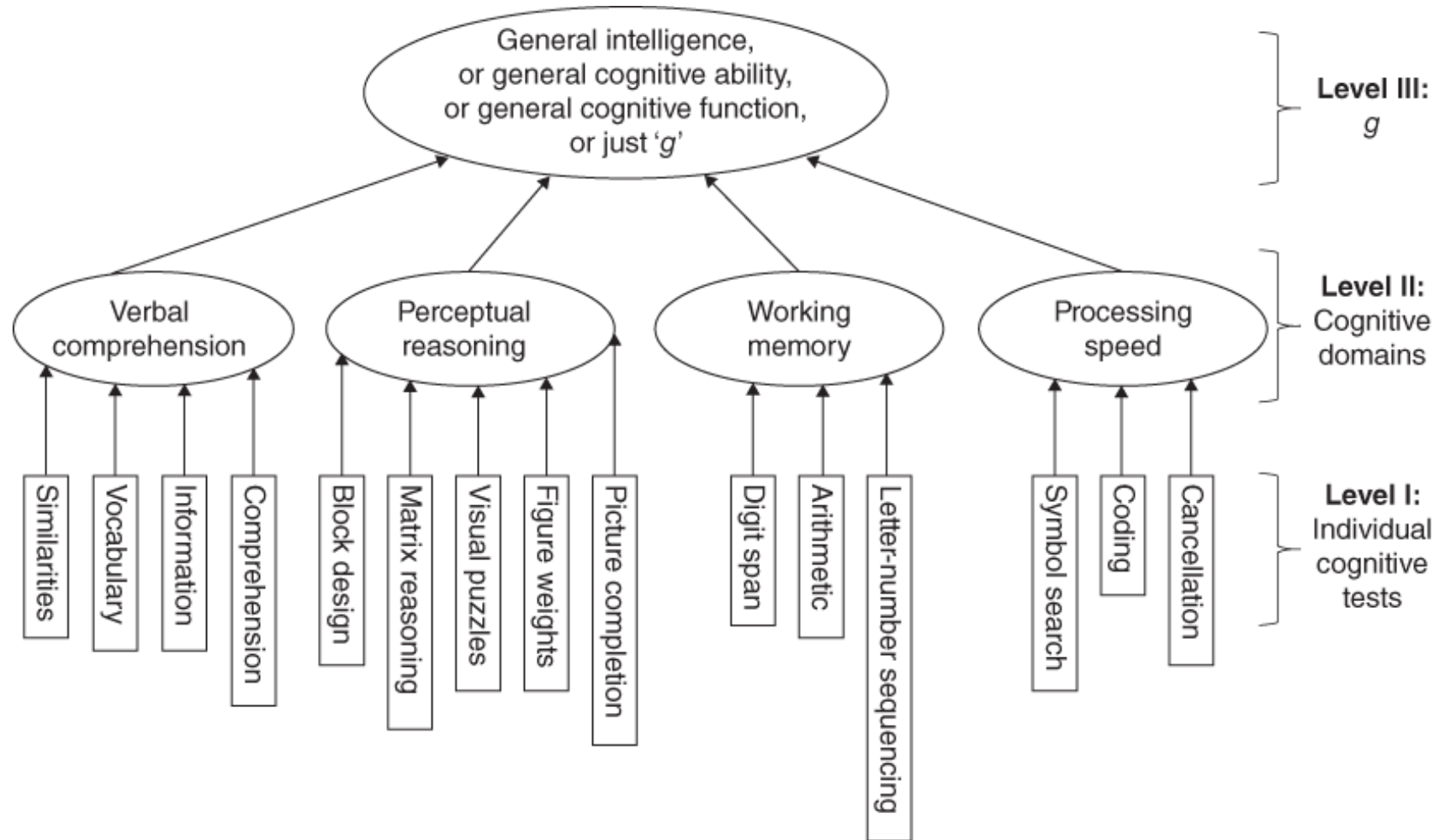
Would being excellent in some tasks carry a price in being poor at others, i.e. a negative correlation between some tests?

The three-level hierarchy of mental ability test scores from the Wechsler Adult Intelligence Scale IV

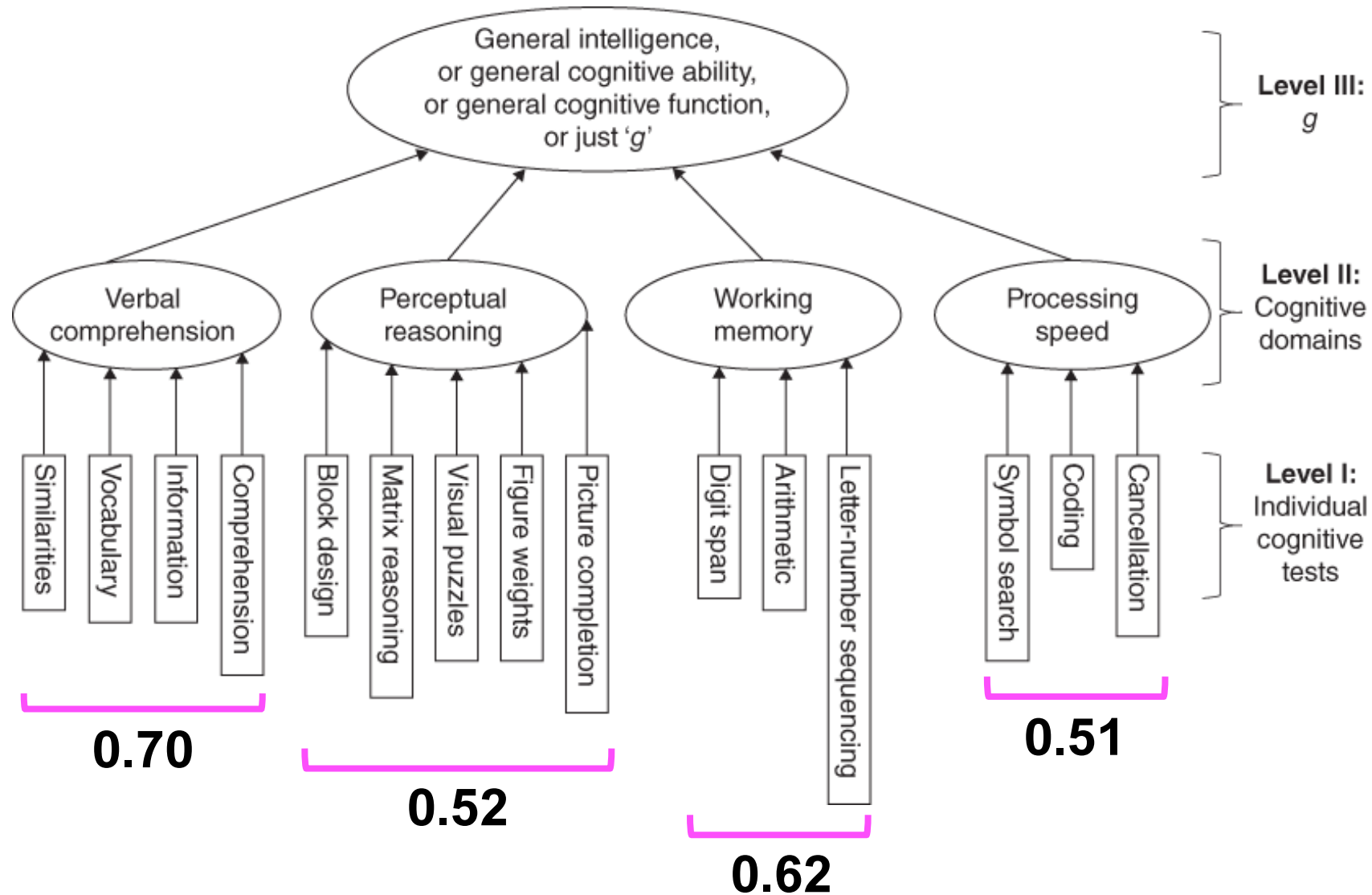


Out of 105 correlations – all were positive, no tests were unrelated to any other!!!
i.e. there were no near-to-zero correlations (data from 1800 ppl. aged 16-69 years)

The three-level hierarchy of mental ability test scores from the Wechsler Adult Intelligence Scale IV



The three-level hierarchy of mental ability test scores from the Wechsler Adult Intelligence Scale IV



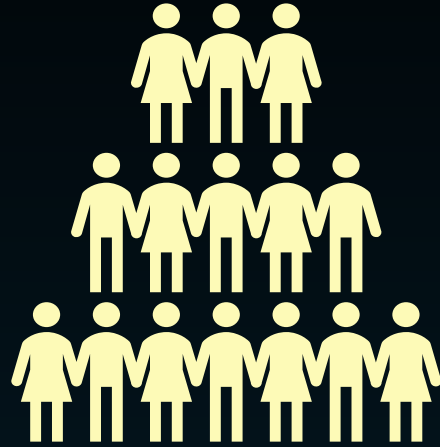
SUMMARY: The Wechsler Intelligence Tests

- Designed to assess intelligence **for all people and all ages**
- Within each subtest there are a variety of items with a wide range of difficulty
- All participants tested on the same items. Only 3 of 15 tests require pencil
- **Deviation IQ** - Wechsler carefully stratified by age as well as by other standard demographic characteristics (sex, social class, region of the country), so an IQ score on WAIS is simply a reflection of an individual's relative standing with respect to others within the same cohort
- 15 varied tests given to 1800 adults found **people who were good at one of the 15 tests tended to be good at the other 14**. Four sub-groups of tests (verbal comprehension, perceptual reasoning, working memory and processing speed) related more highly to each other than the other 3 pools

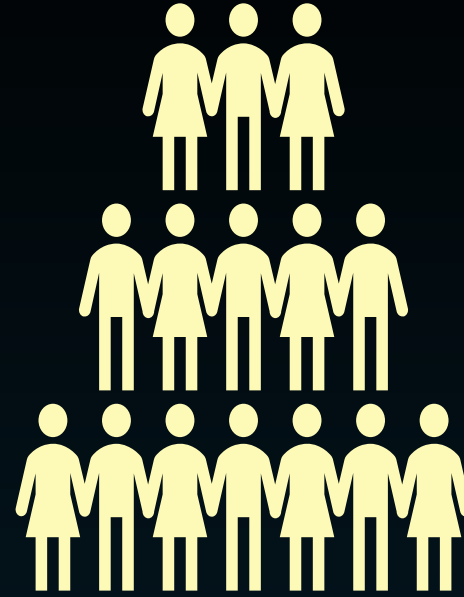
The Wechsler Intelligence Tests



Kesha
12-years-old
IQ of 120



Scores above
average relative
to her peers
(age, sex, social
class, region
etc)



Scores below
average relative
to her peers
(age, sex, social
class, region
etc)



Sherri
23-years-old
IQ of 90

The Wechsler Intelligence Tests

Wechsler Adult Intelligence Scale (WAIS) IQ formula:

$$(\text{actual test score} \div \text{expected test score for that age}) \times 100$$

> All ranges for intelligence scores for all ages were transformed so they had a middle score of 100

The background is a dark blue gradient with a subtle grid pattern. Several concentric circles are visible, some in white and some in a lighter blue. A thick white arc is prominent on the right side, curving from the bottom towards the center. The overall aesthetic is futuristic and technical.

Raven's Progressive Matrices

Raven's Progressive Matrices



John Carlyle Raven
1902-1970

- Standard Progressive Matrices first published 1938
- Non-verbal approach. Identify missing elements in pattern (in the form of matrices)
- Paper and pencil, multiple choice test (large groups can be tested at same time). 60 items increasing in difficulty in each of the 5 blocks
- Freer of cultural, language differences, and better for deaf (minimum of instructions)
- Focussed on one class of test believed to **measure abstract reasoning** and nothing else. Very different from Wechsler tests.
- **Eductive ability** (Latin “educere” meaning to “to draw out”) – think clearly, make sense of complexity
- **Reproductive ability** – to store and reproduce ability

Raven's Progressive Matrices

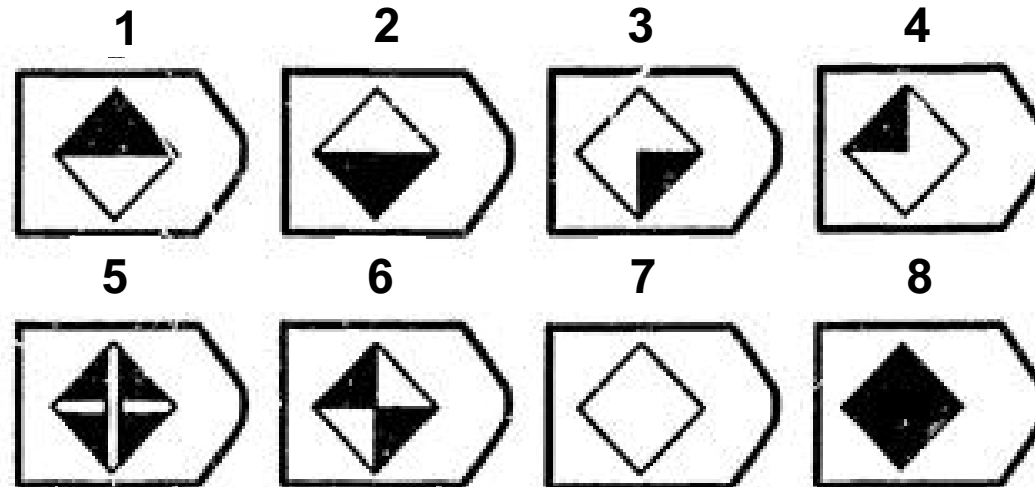
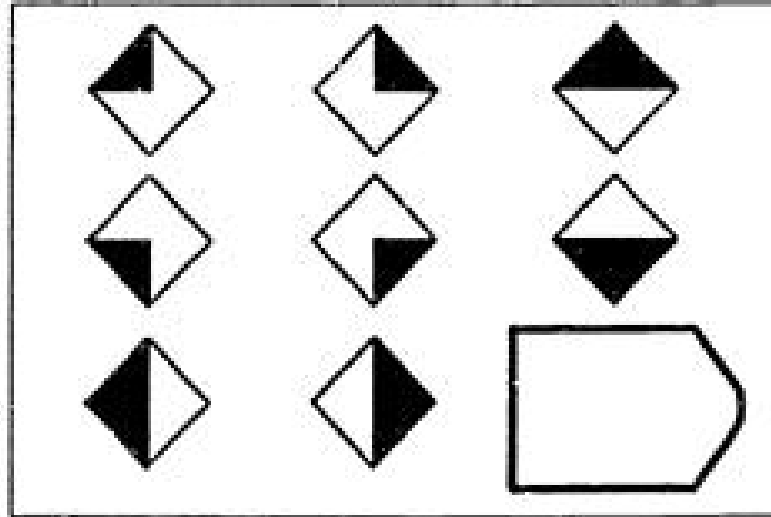


John Carlyle Raven
1902-1970

- Very different from WAIS – yet still the correlations between WAIS and Raven's between 0.40 and 0.75 (Burke, 1958, 1985; Court and Raven, 1995)

Raven's Progressive Matrices

Figuring out the **underlying rules** that explain the progression of shapes.



SUMMARY: Raven's Progressive Matrices

- Designed to tap into **abstract reasoning** (not applied or practical) to see relationships between objects, events and information – this abstraction is theoretically supposed to underlie the g-factor (general intelligence)
- Making meaning out of confusion – useful marker for how well someone can mentally visualise a solution to a visual problem
- Only one class of test, i.e. **not** designed to measure different aspects of intelligence - correlates highly with WAIS despite largely different content
- Testing knowledge not explicitly taught in school, across cultures etc
- Can test 5-year-olds to advanced adults
- Widespread practical application (can be done online) – military conscripts across the world, high IQ societies, job recruitment, educational & clinical settings

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Validity of intelligence tests

Validity of intelligence tests

Validity refers to the question of whether the test measures what it claims to measure.

- ***Face validity***

An IQ test is said to have face validity if it "looks like" it is going to measure what it is supposed to measure.

- ***Concurrent validity***

Intelligence test shows concurrent validity if it shows a (positive) relationship with other measures of intelligence.

- ***Predictive validity***

Intelligence tests have been used to predict 'real-world' measures of intelligence or achievement (e.g. school achievement, job performance, etc.)

Validity of intelligence tests

Reliability refers to two aspects:

Internal reliability

Any measure of intelligence with good internal reliability will have a number of items that correlate positively with one other, this suggests they are measures of the same construct.

Test-retest reliability

A good intelligence test will show a good level of reliability over time. Your general intelligence is thought to be relatively stable over time. You would expect IQ scores to be very similar across test sessions.

Use and misuse of IQ tests

DO IQ TESTS WORK?



TEDEd

<https://www.youtube.com/watch?v=W2bKaw2AJxs>

Use and misuse of IQ tests

Use

Objective way to identify children requiring special education services / or gifted children

Identify structural inequalities affecting development

To assess effectiveness of interventions (e.g. neurofeedback training for ADHD)

Misuse

Exclusion and control of marginalised groups (eugenics, sterilisation, elitism, racism)

Cultural specificity makes tests biased

Treating IQ as a fixed label or single measure of a person's worth.

SUMMARY

- Two major measurements of intelligence (both tests thought to measure 'g'):
 1. Wechsler Adult Intelligence Scale (WAIS) /Wechsler Intelligence Scale for Children (WISC)
 2. Raven's Progressive Matrices
- Two quite different IQ tests, constructed on different principles, administered in different ways, and containing entirely different kinds of items but they still correlate surprisingly highly with one another
- Tests used for clinical assessment (brain injury / dysfunction, diagnostics, rehabilitation), employment screening, education etc
- People who are good at one intelligence test tend to be good at others, regardless of the type of test
- Dark history of intelligence tests – misused as justification for political / ideological gains

END